

08 • L3 — Machine-readable Layer: Architecture & Schema

Assignment: “Develop the machine-readable code (L3) — translate the L2 into an L3 ... Consider user experience (e.g. sequence and grouping of your questions), maintainability of your code and data quality (e.g. constrain rules).”

8.1 The Architecture Decision

We author the L3 in our own, clinically readable JSON format and speak standards at the interface.

Lecture Day 3 explicitly calls for this discussion: “Even aspects that could sound straightforward such as having a L3 based on software neutral standard should be discussed.” Here is our rationale.

Options Compared

Option	Advantages	Disadvantages	Decision
XLSForm / ODK	Established standard, executable by existing engines (ODK Collect, Enketo), covered in the course	Standard widgets represent our core interaction (large-format image cards for laypersons, traffic-light presentation of results) only to a limited extent — the L4 would have dictated the UX	✗
FHIR Questionnaire + CQL	Full interoperability standard, WHO SMART compliant, future-proof	“FHIR resources are not human readable, making review and maintenance challenging” (slide, Day 3). In a 6-day project without FHIR expertise, not reviewable by the clinically responsible person	✗
Hardcoding in the app	Fastest route to a prototype	Explicitly named by the assignment as the inferior option; every rule change requires a developer plus a rebuild	✗
Custom JSON + FHIR at the interface	Reads almost like the L2 — every rule is one line, clinically reviewable. Full control over the UX. Standards where they count: in data exchange	No ecosystem of existing engines can execute it directly. The expression language is JS-like rather than CQL	✓

Why This Is Legitimate

The lecture explicitly permits this route:

“Authored L3 could be tailored for human authoring while not being fully machine-readable [standard]” “L3 might be used as-is in the L4, **or might be translated** to match the target application”

That is exactly what we do: **authoring in the simple format, translation into the standard at the boundary**. Conceptually, our L3 corresponds one to one to a FHIR Questionnaire :

Our L3	FHIR Questionnaire
elements[]	item[]
relevant	enableWhen

Our L3	FHIR Questionnaire
options[]	answerOption[]
constraint	item.extension[maxValue/minValue]
sections[]	item[type=group]

Limits We Disclose

- No existing CDSS ecosystem can execute our L3 directly. An automatic conversion to FHIR Questionnaire would be a possible next step.
- Our expression language is not CQL. For a national rollout the WHO SMART route (FHIR + CQL) would be the right one — for a prototype it would have been overengineering.
- The software neutrality that Day 3 names as a target criterion is **not** something we achieve fully. This is a deliberate trade-off in favour of maintainability and available skills.

One sentence for the slide:

"We author in simple, clinically reviewable JSON and speak standards at the interface — a deliberate trade-off between software neutrality and maintainability, discussed as required in the lecture."

8.2 Assessment Against the L3 Requirements from Day 3

Requirement	WundCheck L3	Status
Structured	JSON schema: meta / sections / elements / decision_rules	✓
Contains workflow rules	relevant , calculation , constraint , when rules in worst-first order	✓
Machine readable	JSON, consumed directly by the L4; L3 can be swapped without a code change	✓
Software neutral	Proprietary format — not executable by ODK/CHT/FHIR engines	⚠ deliberate trade-off
Rely on terminology	SNOMED CT / LOINC / ICD-10 bindings per data element (21 of 48)	✓ partly
Use interoperability standards	Export as FHIR R4 QuestionnaireResponse + Observation + Communication	✓ at the interface

8.3 Schema

```
{
  "meta": {
    "title": "WundCheck – Daily wound check after surgery",
    "version": "0.3.3",
    "languages": ["en", "de"],
    "default_language": "en",
    "language": "de",
    "ll_sources": [ /* CDC/NHSN, NICE NG125, Southampton, Campwala et al. */ ],
    "disclaimer": "Study prototype FHNW MSc MI – not for clinical use.",
    "thresholds": {
      "fever_orange": 38.0,
      "fever_red": 39.0,
    }
  }
}
```

```

    "pain_high":          7,
    "normal_healing_days": 3,
    "discharge_prolonged_days": 3,
    "bmi_risk":           30
  },
  "changelog": [ { "version": "...", "date": "...", "changes": "..." } ]
},

"sections": [
  { "id": "stamm",      "title": "1 · Details of the operation (once)" },
  { "id": "risiko",    "title": "2 · Risk factors (once)" },
  { "id": "allgemein", "title": "3 · Daily check – general condition" },
  { "id": "wunde",     "title": "4 · Daily check – wound" }
],

"elements": [
  {
    "id":          "temp",
    "de_id":       "DE-020",
    "section":     "allgemein",
    "type":        "decimal",
    "label":       { "en": "Body temperature today",
                    "de": "Körpertemperatur heute" },
    "required":    true,
    "constraint":  { "expr": "temp >= 34 and temp <= 43",
                    "message": { "en": "Plausible temperature: 34–43 °C.",
                                "de": "Plausible Temperatur: 34–43 °C." } },
    "terminology": { "loinc": "8310-5" },
    "fhir":        "Observation.valueQuantity",
    "ll_source":   "CDC/NHSN – deep incisional SSI, fever >38 °C"
  },
  {
    "id":          "fever",
    "de_id":       "DE-021",
    "type":        "calculated",
    "calculation": "temp >= meta.thresholds.fever_orange",
    "label":       "Fever"
  },
  {
    "id":          "discharge_extent",
    "de_id":       "DE-037",
    "type":        "select_one",
    "relevant":    "secretion >= 1",
    "label":       "Is the fluid coming from a small spot or along the whole wound?",
    "options":     [ { "value": 1, "label": "Small spot", "image": "assets/..." },
                    { "value": 2, "label": "Along the whole wound" } ]
  }
],

"decision_rules": [
  {
    "id":          "R1",
    "decision":     "Wundstatus",
    "priority":     "red",
    "exclusive":    true,
    "when":         "temp >= meta.thresholds.fever_red or chills = 1",
    "output":       "EMERGENCY – High fever or chills: ... (NOTFALL – Hohes Fieber oder Schüttelfrost: ...)",
  }
]

```

```

      "annotation": "Systemic signs of infection have the highest priority (worst-
first).",
      "reference": "CDC/NHSN SSI · sepsis red flags"
    }
  ]
}

```

Field Semantics

Field	Meaning
<code>de_id / id</code>	Traceability ID, identical to the Data Dictionary
<code>relevant</code>	Visibility condition — controls the conditional logic
<code>calculation</code>	Only for <code>type: calculated</code> ; defines the derived element
<code>constraint</code>	Plausibility check with an error message understandable by laypersons; runs before rule evaluation
<code>terminology</code>	Code bindings (SNOMED CT / LOINC / ICD-10)
<code>fhir</code>	Target resource and element for the export
<code>l1_source</code>	Back-reference to the passage in the guideline
<i>label objects</i>	Every user-facing string is an object keyed by language code ({ "en": ..., "de": ... }). <code>meta.languages</code> lists the available languages, <code>meta.default_language</code> the default. Adding a language means adding a key — no code change
<code>exclusive</code>	<code>true</code> = part of the worst-first cascade · <code>false</code> = informational rule, fires in addition

8.4 Naming Conventions

Prefix / pattern	Use	Examples
<code>pat_*</code>	Patient master data	<code>pat_name</code> , <code>pat_dob</code> , <code>pat_age</code>
<code>op_*</code>	Details of the procedure	<code>op_date</code> , <code>op_site</code> , <code>op_duration_min</code>
<code>wound_*</code>	Wound findings	<code>wound_state</code> , <code>wound_open</code> , <code>wound_class</code>
<code>R1...R12</code>	Classification rules (exclusive)	worst-first
<code>I1...I5</code>	Informational rules (non-exclusive)	fire in addition
<code>DE-xxx</code>	Data element traceability	Data Dictionary ↔ L3 ↔ test

Snake_case throughout. Every ID in the L3 is identical to the ID in the Data Dictionary — this is **binding**, because otherwise traceability breaks.

8.5 The L3 Implementation Package

Day 3 defines the L3 more broadly than just the form file. Mapping of our artefacts:

Package component (Day 3)	Our artefact
Documentation — Process, Persona, Use Cases	01 Health Need · 02 Personas · 04 BPMN
Content / Workflow — Sequencing, Logic Library	<code>l3/wundcheck-l3.json</code> — <code>sections</code> , <code>relevant</code> , <code>decision_rules</code>

Package component (Day 3)	Our artefact
Terminology — ad-hoc concepts, external code systems	Terminology columns in the Data Dictionary and in the L3
Validation — Tests, test data, Conformance Rules	10 QA Testing · test cases T1-T18
Interfaces / Interoperability — Standard, Mapping, Transformation	FHIR export (<code>QuestionnaireResponse</code> + <code>Observation</code> + <code>Communication</code>)
UI/UX — Interfaces, Translations	Image answer options, traffic-light presentation, grouping into sections (09 L4)

8.6 Versioning & Update Management

Lecture Day 3: "Navigating L3 updates"

- The L3 carries a **version number** (`meta.version`) and a **changelog** (`meta.changelog`). The app displays the loaded version in the header.
- The FHIR export references the version (`questionnaire = "WundCheck-L3-v0.3.3"`) — this makes it traceable **which version of the rules** produced a recommendation. In the event of an incident, this is the decisive piece of information.

Downstream Effects of Changes

Change	What additionally has to be adapted
Data element removed	All <code>relevant</code> , <code>calculation</code> and <code>when</code> expressions that reference it
Data element added	Data Dictionary, possibly new rules, possibly new test cases
Threshold changed	Only <code>meta.thresholds</code> — but : full regression test T1-T18
Rule added	Determine its position in the worst-first cascade, add a test case

Change Process

```

Guideline update
└─> Adaptation in the L2 (Data Dictionary / Decision Table)
    └─> Review by the clinically responsible person
        └─> Transfer into the L3 JSON
            └─> Regression test T1-T18
                └─> Increment version + changelog
                    └─> Release

```

This is the core of our maintainability argument: a content change requires **no** developer and **no** rebuild. Only the step "transfer into the L3 JSON" touches a technical file — and that file is deliberately built so that a clinical professional with basic knowledge can read and change it.

8.7 What Lives in the L3 — and Why That Is Testable

Everything that could conceivably be called clinical or configurational sits in the L3:

In the L3	Consequence
Questions, answer options, images, visibility conditions	A new question is a JSON entry
Thresholds (<code>meta.thresholds</code>) with their sources	A guideline change is one line
Decision rules with priority, annotation and L1 reference	A new rule is a JSON entry
Terminology bindings and the code system URIs (<code>meta.terminology_systems</code>)	Switching from SNOMED to a national code system is configuration
FHIR endpoint, patient reference, canonical, alert recipient (<code>meta.fhir</code>)	Redirecting to a hospital server is one line
<code>Observation.code</code> , alert prefix, Observation components	The interface payload is declared, not coded
Which extras appear on the result badge (<code>meta.display</code>)	Presentation of derived values is declared
Which sections are captured once (<code>once</code> flag)	The app does not know which section is clinical

The L4 knows only *types* (`select_one` , `calculated` , `date`), *structure* (sections, elements, rules) and *mechanics* (worst-first, two-pass, constraints before evaluation).

And this is enforced, not merely claimed. `tools/check-hardcoding.mjs` scans the application source for every element id, section id, rule id, threshold, code system URI, the endpoint and a clinical vocabulary list, and fails if any of them appears. See 10 §10.6 — the check found two genuine leaks on its first run.

8.8 Multilingualism

Every user-facing string in the L3 — labels, answer options, constraint messages, rule outputs, section titles, scope and disclaimer — is a language-keyed object:

```
"label": { "en": "Does a red streak run away from the wound?",
           "de": "Zieht ein roter Streifen von der Wunde weg?" }
```

`meta.languages`: [`"en"` , `"de"`] declares which languages exist, `meta.default_language`: `"en"` sets the default. The app renders a language switch from this list and falls back to `en` for any missing key.

A further language is a pure content change: add the key to each string, add the code to `meta.languages` . No code change, no rebuild — the same argument as for the thresholds. For the target population (patients with language barriers) this is not a nice-to-have but a prerequisite for equitable access, see 12 Impact.

Current status: `en` complete, `de` complete. The clinical annotations, reference fields and `l1_source` remain English only — they are aimed at the maintaining team, not at patients.

8.9 Open Points

- ✓ All new elements (DE-006, DE-008-011, DE-016-018, DE-025, DE-027-028, DE-031-032, DE-037-038, DE-043-049) transferred into the L3
- ✓ `meta.thresholds` introduced; every rule references a named constant
- ✓ `southampton_grade` implemented as `calculated`
- ✓ Rules R6a/R6b, R9-R12 and I5 added
- ✓ `meta.l1_sources` replaced with the four actual sources

- ☒ Bilingual labels (en/de), English as the default
 - ☒ `meta.fhir` — endpoint, canonical, alert recipient, Observation code — declared as configuration
 - ☒ Version `0.3.3` , changelog complete
 - ☐ Automatic conversion to a FHIR `Questionnaire` — the route to full software neutrality, out of scope for the prototype
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Related documents: 05 Data Dictionary · 06 Decision Logic · 09 L4 · 11 Implementation